

HARISH REDDY A

Software Engineer | Python & Backend Developer | AI Applications

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PROFESSIONAL SUMMARY

BCA graduate with hands-on experience designing and shipping backend systems, REST APIs, and data-driven applications using Python, FastAPI, and SQL. Built and deployed a full-stack AI application integrating a Retrieval-Augmented Generation pipeline with PostgreSQL and vector search. Strong fundamentals in database design, authentication, and version-controlled, collaborative development. Comfortable moving from data analysis to production code, with a track record of automating manual workflows and delivering measurable efficiency gains.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, HTML5, CSS3
- **Backend & APIs:** FastAPI, REST APIs, JWT Authentication, SQLAlchemy, Alembic Migrations, Repository-Service Pattern
- **Frontend:** React
- **AI/ML:** LLM Integration, Retrieval-Augmented Generation (RAG), LangChain, pgvector, Prompt Engineering, OCR
- **Databases:** PostgreSQL, MySQL
- **DevOps & Tools:** Git, GitHub, Docker, Swagger/OpenAPI, VS Code

PROFESSIONAL EXPERIENCE

Data Science Intern | Skill Forge E-Learning Solutions Pvt. Ltd., Bengaluru

Jun 2024 – Jul 2024

- Processed and analysed 10,000+ records with a cross-functional team, surfacing operational trends that informed business decisions.
- Built Python automation scripts that cut manual pre-processing effort by 30%, directly improving reporting turnaround time.
- Built web-scraping pipelines to source external market data, growing the dataset available for downstream analysis.
- Wrote and maintained process documentation, keeping analytical work reproducible and audit-ready.

PROJECTS

DevPilot AI – AI Developer Assistant | Python, FastAPI, React, PostgreSQL, SQLAlchemy, pgvector, LangChain, Docker

- Designed and built a full-stack developer-assistant app: FastAPI backend, React frontend, PostgreSQL database, and JWT-based authentication, exposed through versioned REST APIs.
- Built a Retrieval-Augmented Generation pipeline - document parsing, OCR, text chunking, and pgvector embeddings - so the AI could answer questions and generate code from a user's own uploaded files.
- Structured the backend with the Repository-Service pattern and SQLAlchemy ORM with Alembic migrations, keeping the codebase testable and easy to extend.
- Containerized the app with Docker and documented all endpoints with Swagger/OpenAPI for handoff and integration.

Voice Assistant Application | Python, Speech Recognition, NLP

- Built a Python voice assistant that parses spoken commands and triggers the matching action, using speech-to-text and text-to-speech modules.
- Wrote the command-handling logic and debugged edge cases to make recognition more reliable across accents and phrasing.
- Modularized the codebase into separate input, command-processing, and output layers, making it straightforward to add new voice commands.
- Managed the project through Git version control, iterating across multiple builds to improve recognition accuracy and response time.

Coffee Sales Exploratory Data Analysis | Python, Pandas, NumPy, Matplotlib

- Cleaned and analysed a coffee sales dataset end-to-end, from raw data to a documented set of business findings.
- Built visualizations that made seasonal and product-level sales patterns easy to act on.
- Applied statistical analysis to identify top-performing products and peak sales periods, supporting inventory and promotion decisions.
- Documented the full analysis workflow in a reproducible notebook format for handoff and future reuse.

EDUCATION

Bachelor of Computer Application (BCA)

Bengaluru North University

2021 – 2024

CGPA: 9.36/10

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Python for Data Science -HCL
- Google Cloud Digital Leader
- Tata GenAI Powered Data Analytics – Job Simulation (Forage)